Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

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Abstract

Climate-driven shifts in stock productivity reach fishing fleets unevenly when species portfolios differ across segments and the sector-level allocation of catch rights cannot adjust to relative productivity changes. We quantify this distributional channel for the small pelagic fishery of Central-South Chile (anchoveta, sardine, jack mackerel), managed under a fixed industrial–artisanal allocation. We couple a Bayesian state-space model of stock dynamics, fit to 2000–2024 official assessment data, with a vessel-level trip equation; climate enters through an indirect biomass channel and a direct weather channel. Under a six-model CMIP6 ensemble, projected fleet-level effort falls by 13–15 percent for the artisanal fleet but only 0.8–1.0 percent for the industrial fleet—an asymmetry of roughly fifteen to one driven by differential portfolio exposure and weather sensitivity. The findings imply that climate-adaptation policy should revisit the sector-level allocation of catch rights, allowing it to respond to relative productivity shifts.

Keywords: Bayesian state-space model; Bioeconomic projection; Chilean small pelagic fishery; Climate change; Fishing effort; Fleet heterogeneity; Quota allocation

JEL Codes: Q22, Q54, Q57, Q58

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Introduction

Climate change is reshaping the productivity of marine fisheries, but the incidence of this reshaping across fleet segments is rarely uniform. Quota-allocation regimes that fix the industrial–artisanal split of the catch were typically designed under stationary-climate assumptions, so a fleet-asymmetric productivity shock acts as a de facto reallocation of harvest rights. Whether such regimes should anticipate this reallocation depends on the projected magnitude and asymmetry of fleet-level effort responses under credible climate scenarios—a quantity that the empirical literature has rarely delivered jointly for multi-species fisheries managed under sector-asymmetric allocation regimes.

We provide that quantity for the small pelagic fishery (SPF) of Central-South Chile (from Valparaíso to Los Lagos), part of the country's largest fishery—approximately 86% of national fish landings by industrial and artisanal sectors in 2019 (SUBPESCA 2020)—and the segment in which all three target species, anchoveta (*Engraulis ringens*), sardine (*Strangomera bentincki*; locally, sardina común), and jack mackerel (*Trachurus murphyi*; locally, jurel), coexist in non-negligible shares of both fleet portfolios. These three species share habitat and are coupled through trophic and market linkages (Alheit and Niquen 2004; Arancibia et al. 2019; Dresdner et al. 2013). The fishery is managed through a sector-specific quota architecture: a fixed industrial-versus-artisanal allocation splits the Total Allowable Catch between the two segments, while within-sector distribution operates through Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) for the industrial fleet and the Artisanal Extraction Regime (*Régimen Artesanal de Extracción*, RAE) for the artisanal fleet, with limited cross-sector transferability. Because the two segments target different species portfolios, a stock-asymmetric climate shock translates directly into a fleet-asymmetric effort response. Cross-country evidence indicates that Chile is projected to lose part of its maximum catch potential under climate change: Cheung et al. (2010) classify Chile among a group of countries with moderate projected declines (6–13% by 2055), including the contiguous United States, China and Brazil; the available climate-fisheries economics literature is patchy globally and concentrated in a few well-studied regions (Sumaila et al. 2011). To our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.[1]

Our bioeconomic framework combines two empirical components. First, a Bayesian state-space model of multi-species stock dynamics — fitted to 2000–2024 official assessments from the Chilean Fisheries Development Institute (Instituto de Fomento Pesquero, IFOP) and the South Pacific Regional Fisheries Management Organisation (SPRFMO) — estimates, for each species, two semi-elasticities of productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies. We call these parameters *climate shifters*. Because they enter the biological law of motion as coefficients on exogenous climate forcings, they retain a structural interpretation when applied to climate states outside the 2000–2024 sample. Second, a vessel-level negative binomial trip equation — estimated separately for the industrial and artisanal fleets, with year fixed effects — maps prices, allocated harvest, weather, and regulatory closures into annual fishing effort. Climate reaches fleet-level effort through two channels: an *indirect biomass channel* via the climate shifters, and a *direct weather channel* via vessel-specific exposure to severe winds. We evaluate the framework against a six-model ensemble from the Coupled Model Intercomparison Project Phase 6 (CMIP6) under two Shared Socioeconomic Pathway scenarios, SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions), at mid-century (2041–2060) and end-of-century (2081–2100).

We obtain three sets of results. First, the artisanal segment absorbs most of the projected climate impact: fleet-level effort falls by 13–15% for artisanal vessels but only 0.8–1.0% for industrial vessels under SSP5-8.5, an asymmetry of roughly fifteen to one. Second, this asymmetry is driven by differential portfolio exposure combined with limited cross-sector transferability: the artisanal fleet is concentrated in anchoveta and sardine, whose climate shifters are large and well-identified, while 95% of the industrial fleet's portfolio sits in jack mackerel, whose Centro-Sur climate shifter the data cannot identify. Third, the within-artisanal heterogeneity is itself first-order: the dominant *lancha motor* segment contracts by -19.9%, the smaller *bote motor* segment by only -3.2%.

The remainder of the paper is organized as follows. The next section describes the economic and institutional setting of the Chilean SPF. We then present the empirical framework — data, stock-dynamics model, trip equation, and projection design — followed by results on identified climate shifters, comparative-statics projections of stock productivity, and projections of fleet-level effort.

[1] The only empirical study of Chilean fisher behavior that we are aware of is Peña-Torres et al. (2017), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

The paper closes with a discussion of policy implications and limitations, and brief concluding remarks.

The small pelagic fishery in Chile

The fishery is divided into two latitudinal zones with distinct species compositions. The northern zone (from Arica y Parinacota to Coquimbo) is dominated by anchoveta and jack mackerel, while sardine is effectively absent. In contrast, the Central-South zone (from Valparaíso to Los Lagos) includes all three species in non-negligible shares within both artisanal and industrial landings. Because this region supports a genuinely multi-species fishery operating under a fixed sector-level allocation regime, it provides the relevant setting for evaluating the distributional consequences of climate-induced shifts in stock productivity, and is therefore the focus of this paper. The principal ports servicing this zone include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. The three species are harvested primarily by purse-seine fleets that differ in vessel size, operating range, and species composition.

The historical evolution of the fishery illustrates the portfolio mechanism developed in the remainder of the paper. The jack mackerel fishery was initially concentrated in northern Chile but shifted toward the Central-South zone during the mid-1980s (Peña-Torres et al. 2017), where fishing grounds have traditionally operated within 50 nautical miles of the coast. Between 1987 and 2004, approximately 85% of jack mackerel landings were directed to the reduction industry (Peña-Torres et al. 2017), reflecting the importance of fishmeal and fish-oil export markets in determining the value of the industrial portfolio. Sardine and anchoveta, by contrast, are jointly harvested using the same gear and fishing grounds. The two species share habitat and fishing technology, making species differentiation at the point of capture effectively impossible (Dresdner et al. 2013). As a result, the binding constraint on the species mix of landings is the quota regime, not biology or technology.

The status of the three stocks has evolved substantially over time. Central-South anchoveta was classified as collapsed through 2018, reclassified as overexploited in 2019, and has been harvested within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within

MSY reference levels, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel remained overexploited until 2018 and has since been harvested within MSY limits.

Biological closures (*vedas*) constitute a central management instrument for sardine and anchoveta. In the Central-South zone, reproductive closures extend from July to October in Araucanía–Los Lagos and from August to October in Valparaíso–Biobío, while recruitment closures occur from January through February across all regions. In total, these regulations generate 151 closed fishing days per year in Valparaíso–Biobío and 182 days in Araucanía–Los Lagos. Jack mackerel is not subject to seasonal biological closures.

The Chilean SPF is managed through an annual Total Allowable Catch (TAC; *Cuota Global*), further subdivided across regions and seasons, with unused quotas reassignable within the fishing year and small fractions reserved for research and contingency. Anchoveta and sardine operate under a mixed-species quota framework in which each species has an independent quota but substitution between them is permitted.

The partition of each TAC between the industrial and artisanal sectors is fixed by statute. The pre-reform partition governed the de jure allocation during the estimation window (2013–2024); a 2025 reform (Congreso Nacional de Chile 2025) fixes a new partition in force until 2040 (Table 1). Under the pre-reform regime, the artisanal sector held 78% of the Central-South anchoveta and sardine TACs and 10% of the jack mackerel TAC; the 2025 reform raises the anchoveta and sardine artisanal shares to 90% across Valparaíso–Los Lagos, and the jack mackerel artisanal share to 30% in Valparaíso–Los Ríos and 15% in Los Lagos. The de jure partition admits a single administratively rationed inter-sectoral re-allocation channel, subject to ex-ante authorisation by the Subsecretaría de Pesca y Acuicultura (SUBPESCA). Quota control reports from the Servicio Nacional de Pesca y Acuicultura (SERNAPESCA) for 2013–2024 document that the industrial sector cedes a persistently large share of its assigned anchoveta and sardine TACs to artisanal cessionaries — from 12–72% over 2013–2017, tightening to 85–99% every year in 2018–2024 (Online Appendix, Table H.3) — while jack mackerel cessions are an order of magnitude smaller and shift direction across years (Online Appendix, Table H.4). Our empirical specifications are therefore identified on realised landings, which are post-cession: the post-cession allocation reflects the joint statutory–cession regime, not the de jure fractioning alone.

Within the industrial sector, an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) has operated since 2013: Class A licenses were allocated according to historical catch shares, whereas Class B licenses—up to 15% of the industrial allocation—have been distributed through sealed-bid first-price auctions since 2015.[2] The artisanal sector operates under a limited-entry regime, with regional TACs distributed through the *Régimen Artesanal de Extracción* (RAE) according to area, vessel size, landing site, or fisher organization.

Table 1: Statutory sectoral partition of the Central-South small pelagic TACs, pre- and post-2025 reform.

	Artisanal share of TAC (%)	
Stock	Pre-reform	2025 reform
Anchoveta (Valparaíso–Los Lagos)	78	90
Sardina común (Valparaíso–Los Lagos)	78	90
Jack mackerel (Valparaíso–Los Ríos)	10	30
Jack mackerel (Los Lagos)	10	15

Notes. Sources: SERNAPESCA (2024) for the pre-2025 regime and Congreso Nacional de Chile (2025) for the 2025 reform, in force until 31 December 2040.

Methods

Our empirical strategy targets two structural objects—climate shifters of stock productivity and fleet-specific elasticities of fishing effort—and combines them to compute counterfactual fleet responses under future climate paths. First, we identify the climate shifters: species-specific semi-elasticities of intrinsic stock productivity with respect to sea-surface temperature and chlorophyll-a anomalies. We estimate them within a Bayesian state-space model of multi-species stock dynamics, with a Schaefer law of motion and priors taken from official IFOP and SPRFMO assessments. Second, we estimate fleet-specific elasticities of annual fishing effort with respect to allocated harvest,

[2] Peña-Torres et al. (2022) document that these Class B auctions have exhibited low participation, limited capacity to reflect economies of scale, and indications of coordinated bidding.

ex-vessel prices, weather exposure, and regulatory closures, using a negative binomial count model fitted separately for the artisanal and industrial fleets, with year fixed effects that absorb aggregate non-climate shocks. Third, we compute long-run comparative statics under a six-model CMIP6 ensemble (SSP2-4.5 and SSP5-8.5, mid- and end-of-century), propagating the climate shifters through the stock-dynamics steady state to long-run productivity and through the trip equation to fleet-level effort.

Historical data

We use data requested from IFOP covering the 2013–2024 period. The dataset includes trip-level microdata with detailed records on vessel identifiers, departure and arrival times, vessel capacity, fleet and vessel type, ports of departure and landing, fishery codes, haul timing and location, species composition, and retained catch. We also requested annual stock biomass series for 2000–2024 from official IFOP and SPRFMO assessments and aggregate landings by region and species from SERNAPESCA. For prices, we use ex-vessel monthly series from IFOP's *Monitoreo Económico de la Industria Pesquera y Acuícola Nacional*, which records monthly raw-material payments by processing plants to fishers, disaggregated by plant, region, species, raw-material type, and consumption class (human or animal). Processing-volume series are reported on a census basis across plants, while the price series is sample-based.

Environmental covariates are obtained from the E.U. Copernicus Marine Service Information, accessed through the Copernicus Marine Toolbox API. Sea surface temperature comes from the Global Ocean Physics Reanalysis (GLORYS12V1) at 1/12° horizontal resolution (E.U. Copernicus Marine Service Information 2025c). Surface wind speed is taken from the Global Ocean Hourly Reprocessed Sea Surface Wind and Stress from Scatterometer and Model dataset, at 0.125° horizontal resolution and hourly frequency (E.U. Copernicus Marine Service Information 2025b). Chlorophyll-a concentrations come from the Global Ocean Colour dataset at $\sim$ 4~km horizontal resolution (E.U. Copernicus Marine Service Information 2025a). All environmental data were retrieved daily (hourly for winds) for the 2013–2024 period, covering the Chilean Exclusive Economic Zone (EEZ) between 32°S and 42°S (Figure 1).

Sample structure for jack mackerel biomass identification The jack mackerel biomass sample over the 2000–2024 estimation window is informative for identification in 18 of 25 years. Sixteen years contribute a point estimate from the official IFOP hydroacoustic assessment, and two additional years (2012, 2015) enter the likelihood as left-censored observations at the assessment's lower detection limit. The remaining seven years carry no biomass observation; their latent state is identified dynamically by the law of motion of Section , together with the priors on $(r^0, K, \sigma_{\text{proc}}, \sigma_{\text{obs}})$ from official IFOP and SPRFMO assessments and on (ρ^{SST}, ρ^{CHL}) from the reduced-form stress tests of Online Appendix A. This reduced informational content is one of the load-bearing facts behind the non-identification result for the jack mackerel climate shifter (Section).

Two external series—jack mackerel biomass from the Northern acoustic zone (Arica–Iquique–Antofagasta, available since 2010) and the SPRFMO transzonal spawning biomass index for the Southeast Pacific—were considered as auxiliary information. The Northern series correlates with the Centro-Sur (CS) series at $r = 0.82$ over $N = 7$ overlapping years, while the SPRFMO index correlates with CS at $r = 0.11$ over $N = 17$ overlapping years. We treat these numbers as suggestive rather than conclusive: a seven-point Pearson correlation is at best descriptive, but the contrast in sign and magnitude is consistent with a Centro-Sur stock unit that shares short-run dynamics with the Northern Chilean assessment and is only weakly coupled to the SPRFMO-scale aggregate. Neither external series enters the structural model's likelihood—as a prior or as an observation—to avoid introducing information about coastal climate forcing through the indirect route of Northern or transzonal biomass.

Future climate data

Climate projections for SST, chlorophyll-a, and near-surface wind speed are obtained from the CMIP6 archive using a six-model ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR), selected to span the CMIP6 equilibrium climate sensitivity distribution. We use monthly outputs for two future scenarios: SSP2-4.5 (moderate) and SSP5-8.5 (high emissions). Section *Projection approach* describes the projection horizons, baseline period, and delta-method procedure used to translate CMIP6 outputs into our empirical models. The per-model decomposition of projection uncertainty appears in Online Appendix F for

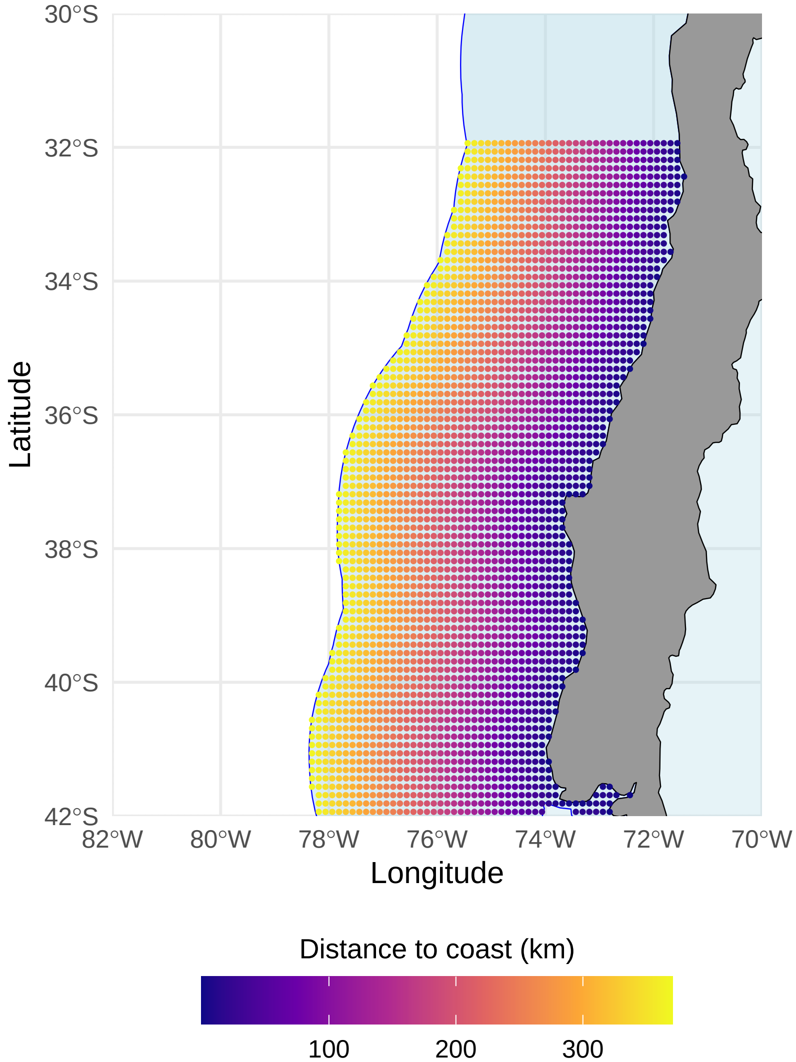

Figure 1: Geographical extent of the study area used for environmental covariates limited to the Chilean Exclusive Economic Zone

stock-level intrinsic productivity and in Online Appendix G for fleet-level trips.

Econometric models

Stock dynamics

Our identification target in this subsection is the vector of climate semi-elasticities $(\rho_i^{SST}, \rho_i^{CHL})_{i=1}^{3}$—the structural parameters that translate sea-surface temperature and chlorophyll-a anomalies into shifts in long-run stock productivity. We recover them by inverting a Bayesian state-space model of multi-species stock dynamics in which a Schaefer transition with a log-linear climate shifter on the intrinsic growth rate governs the latent biomass $B_{i,t}$ of stock $i \in \{\text{anchoveta}, \text{sardine}, \text{jack mackerel}\}$. The Schaefer form (the $\theta = 1$ case of the Pella–Tomlinson family) is adopted on identifiability grounds: with $N = 25$ annual observations per stock, the shape parameter θ is notoriously weakly identified, and imposing $\theta = 1$ is a standard identification device in the small-pelagic literature (Hilborn and Walters 1992; Maunder and Punt 2013). The law of motion is

$$B_{i,t+1} \;=\; B_{i,t} + r_{i,t}\, B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) \;-\; C_{i,t} \;+\; \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{proc},i}), \tag{1}$$

where $C_{i,t}$ is the observed annual catch, K_i is the stock-specific carrying capacity, and $\sigma_{\text{proc},i}$ is the standard deviation of the stochastic recruitment innovation. Climate enters the dynamics through a log-linear shifter on the intrinsic growth rate,

$$r_{i,t} \;=\; r_i^0\, \exp\Big(\rho_i^{SST}\,(SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL}\,(\log CHL_{t-1} - \overline{\log CHL})\Big), \tag{2}$$

so ρ_i^{SST} and ρ_i^{CHL} are the semi-elasticities of intrinsic stock productivity to a one-degree Celsius SST anomaly and to a one-log-point chlorophyll-a anomaly, respectively. Because these parameters enter the law of motion as coefficients on exogenous climate forcings, they are invariant to the joint historical distribution of (SST, B, C)—transportable, in the structural sense, to climate regimes outside the 2000–2024 estimation sample. This transportability matters because the comparative-statics exercise of Section *Climate change projections* evaluates the climate shifter at projected SST anomalies that lie several historical standard deviations beyond the estimation envelope.

Latent biomass is linked to the official survey-based estimates of spawning stock biomass $B_{i,t}^{\text{obs}}$—published by IFOP for anchoveta and sardine in Centro-Sur Chile and by SPRFMO for jack mackerel—through a log-normal measurement equation,

$$\log B_{i,t}^{\text{obs}} \;=\; \log B_{i,t} \;+\; u_{i,t}, \qquad u_{i,t} \sim \mathcal{N}(0,\, \sigma^2_{\text{obs},i}). \tag{3}$$

Environmental conditions are summarized annually by sea surface temperature (SST) and chlorophyll-a concentration (CHL), averaged over the Centro-Sur region within the Chilean EEZ and lagged one year to respect the pre-determined ordering of environmental forcing relative to year-t biomass. The choice of these two covariates follows a large biological literature that identifies thermal regime and primary productivity as the dominant local drivers of small-pelagic dynamics in the Humboldt Current System (Axbard 2016; Cahuin et al. 2009; Yáñez et al. 2014; see Cheung et al. 2008; Jennings et al. 2008).

We estimate the state-space specification by Hamiltonian Monte Carlo in Stan. With $N = 25$ annual observations per stock, joint identification of $(r_i^0, K_i, \rho_i^{SST}, \rho_i^{CHL})$ requires the prior information embedded in the IFOP and SPRFMO single-species assessments; Online Appendix A documents the resulting prior elicitation protocol and the boundary problem under maximum likelihood. We estimate three nested specifications and adopt the climate-augmented version of Eq. (2) as our primary object of interpretation, with the nested predictive comparison relegated to Online Appendix B and convergence diagnostics ($\hat{R} < 1.01$ and bulk/tail ESS > 400 for all top-level parameters) to Online Appendix D. The seven jack mackerel Centro-Sur years without a survey estimate enter the likelihood as latent biomass states identified dynamically; two further years (2012, 2015) enter through a log-normal censored-observation term at the assessment detection limit, with all associated uncertainty propagated directly into the posterior of $(\rho^{SST}_{\text{jack mackerel}}, \rho^{CHL}_{\text{jack mackerel}})$.

The state-space formulation in Eq. (1)–(3) separates process from observation noise, enforces a pre-determined ordering between year-$(t-1)$ environmental forcing and year-t latent biomass, and yields structural climate semi-elasticities that are transportable to climate regimes outside the 2000–2024 estimation window—the property that supports the comparative-statics exercise of Section *Climate change projections* at projected SST anomalies several historical standard deviations beyond the estimation envelope.

Figure 2 shows that anchoveta, sardine, and jack mackerel exhibit distinct biomass trajectories over the 2000–2024 window, with no evidence of strong interannual co-movement across species once scale differences are accounted for. Harvest pressure is visibly associated with these fluctuations; jack mackerel in particular experienced an abrupt decline during the late 2000s, consistent with the combined effects of intense fishing and unfavorable environmental conditions.

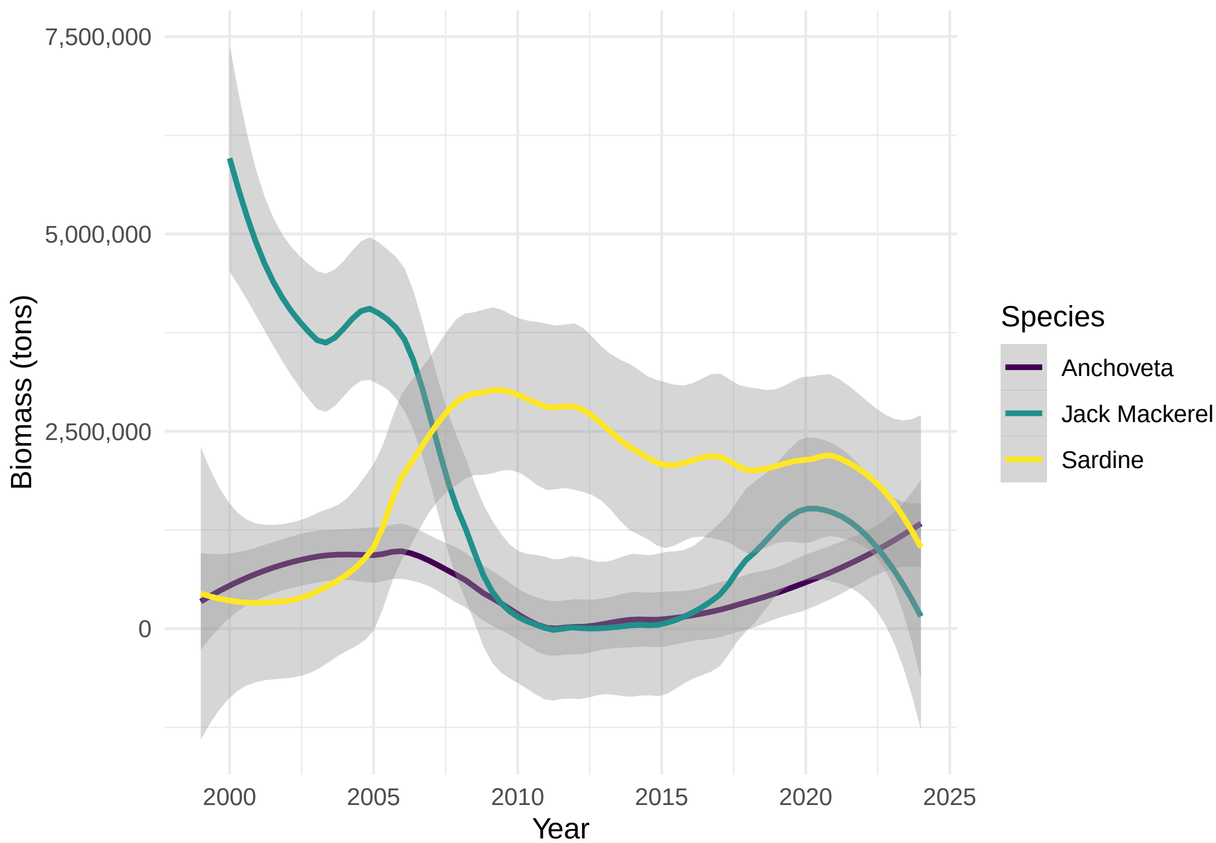

Figure 2: Estimated biomass of small pelagic species in Chile (2000–2024)

Total annual trips

We model the annual number of fishing trips taken by vessel v in year y as a count process following Kasperski (2015). Because the available logbook data record only purse-seine operations, the unit of observation is a vessel–year. We estimate separate effort models for the industrial and artisanal fleets, allowing fishing activity to respond differently to economic conditions and regulatory constraints across sectors. This specification explicitly accommodates technological heterogeneity across fleet segments, reflecting differences in vessel capacity, operating scale, and production technology. A complementary approach, used in Quezada-Escalona et al. (2026) for the U.S. West Coast Coastal Pelagic Species fishery, models daily participation and target-species choice through a discrete choice framework with species distribution models as proxies for daily availability. The annual NB specification we adopt here is the natural counterpart in a setting where vessel-year is the operational unit at which TACs are allocated under the LTP, and where daily logbook coverage is too sparse to support a daily DCM for the Centro-Sur Chilean SPF over the 2013–2024 window.

The baseline specification is a Poisson model:

$$T_{vy} \sim \text{Poisson}(\lambda_{vy}), \qquad \lambda_{vy} = \exp\left(U'_{vy}\beta\right), \tag{4}$$

where T_{vy} denotes the total number of purse-seine trips recorded for vessel v in year y. Given substantial overdispersion in the data—the variance-to-mean ratio of T_{vy} is 22.4 for the artisanal fleet and 4.9 for the industrial fleet—we estimate a negative binomial (NB) model, which nests the Poisson as a special case. A likelihood ratio test strongly rejects the Poisson restriction for both fleets ($p < 0.001$).

The vector of explanatory variables U_{vy} includes output prices by species, vessel-level allocated harvest, fixed vessel characteristics, and operating conditions:

$$U_{vy} = [p_{sy},\, H^{alloc}_{vy},\, Z_v,\, O_{vy}]. \tag{5}$$

Output prices p_{sy} are species-specific ex-vessel prices paid by processing plants to fishers, obtained from IFOP's manufacturing survey and deflated to constant 2018 pesos using the consumer price index from the Central Bank of Chile. Following Dresdner et al. (2013), prices are measured as annual averages over peak fishing months to reflect the economically relevant conditions faced by vessels when planning annual effort.

Because the model aims to characterize vessels' effort responses to changes in prices, quotas, and environmental conditions within the small pelagic fishery, we restrict attention to SPF trips recorded in logbooks. These trips account for approximately 95% of observed fishing revenue and therefore capture the primary economic margin through which vessels adjust annual effort. Cross-validation against the SERNAPESCA vessel-level landings database (transparency request AH010T0006857; 2013–2024 Centro-Sur) confirms that the resulting panel is representative of the purse-seine artisanal segment. For the industrial fleet, the panel covers 40 of 59 registered vessels (68% by count) and aggregate landings within −10% of the SERNAPESCA all-gear total, with portfolio-weighted catch composition matching SERNAPESCA aggregates within two percentage points across the three target species. For the artisanal fleet, the panel covers 859 of 2{,}689 registered vessels (32% by count) but accounts for approximately 70% of total Centro-Sur artisanal landings of the three species, reflecting the concentration of catch in the larger purse-seine vessels for which logbook submission is mandatory. Smaller artisanal vessels using non-purse-seine gears (lampara, lines) operate outside the LTP quota architecture and are not modeled in the trip equation; they appear in the SERNAPESCA aggregate landings used as the likelihood input for the biomass state-space model.

Quota shares do not enter Eq. (4) directly. Instead, shares translate annual TACs into vessel-level allocated harvests. Let ω^r_{vs} denote vessel v's historical share of landings of species s within its administrative region r (for artisanal vessels) or regulatory zone (for industrial vessels), computed over the full sample period. Given the effective TAC $\bar{Q}^r_{sy}$ for species s in year y and region r—obtained from SERNAPESCA quota monitoring records and reflecting all inter-fleet transfers and adjustments—vessel-level allocated harvest is:

$$H^{alloc}_{vy,s} = \omega^r_{vs}\,\bar{Q}^r_{sy}, \qquad H^{alloc}_{vy} = \sum_s H^{alloc}_{vy,s}. \tag{6}$$

Each vessel is assigned to its administrative region based on its modal departure port in the logbook records. Artisanal TACs are assigned at the regional level (Valparaíso, Biobío, Araucanía, Los Ríos, Los Lagos), while industrial TACs follow the regulatory zone structure established by SUBPESCA (the Valparaíso–Araucanía and Los Ríos–Los Lagos zones for jack mackerel; the Valparaíso–Los Lagos zone for sardine and anchoveta). This regional construction introduces cross-sectional variation in H^{alloc}_{vy} that reflects the heterogeneous quota environments faced by vessels operating in different parts of the Centro-Sur fishery.

Following Kasperski (2015), the trip equation enters allocated harvest *by species* rather than as a scalar aggregate: the regressor vector contains $H^{alloc}_{vy,s}$ for $s \in \{\text{anchoveta, sardine, jack mackerel}\}$ with separate semi-elasticities β^s_h. Ex-vessel prices likewise enter at $(year, region)$ resolution, mapped to each vessel through its modal departure region. For the artisanal fleet, the response to allocated anchoveta is allowed to vary by vessel type through an interaction $H^{alloc}_{vy,\text{anchoveta}} \times$ TIPO_EMB, motivated by the observation in the data that small-vessel categories (LM = *lan-*

cha motor, the dominant 77% of the artisanal panel) exhibit a near-zero elasticity to anchoveta quota, while medium-vessel categories (BM = *bote motor*, 8%) exhibit a strong positive elasticity in the spirit of the Kasperski (2015) benchmark. The interaction is estimated only for TIPO_EMB categories with at least 50 vessel-year observations (BM, LM, UNK); three small categories with $N < 50$ (BR = *bote remo*, plus the obsolete codes BRV = *bote remo vela* and L = *lancha*) collectively represent less than 0.1% of artisanal effort and quota and are reported with an aggregated rather than category-specific coefficient. The Kasperski-aligned specification is preferred by an information criterion of ΔAIC = −167.9 for the artisanal fleet (and −5.0 for the industrial fleet) relative to the legacy scalar specification, with multicollinearity diagnostics (VIFs below 2.5 across all six regressors and both fleets) confirming separate identification of the three β_h^s.

The vector Z_v contains time-invariant vessel characteristics, including hold capacity (log of cubic meters) and vessel type. Hold capacity determines the physical constraint on catch per trip and proxies for vessel scale.

The vector O_{vy} captures operating and regulatory constraints that vary across both vessels and years. Unlike the environmental variables in the stock dynamics model, which reflect biological productivity (SST, chlorophyll-a), the variables in the trip equation capture conditions that affect the *feasibility* of fishing operations. Two vessel–year variables are constructed using each vessel's center of gravity (COG)—the catch-weighted centroid of its haul locations over the sample period. First, the number of adverse-weather days is computed as the count of days per year in which wind speed exceeds 8 m/s at the environmental grid point nearest to the vessel's COG.[3] Second, the number of days closed for biological closures (*vedas*) is assigned based on the regulatory zone corresponding to the vessel's COG latitude. In the Centro-Sur region, sardine and anchoveta face seasonal closures for reproduction (August–October in Valparaíso–Biobío; July–October in Araucanía–Los Lagos) and recruitment (January–February in all regions), yielding 151 and 182 closed days per year, respectively. Jack mackerel has no biological closures.

Quota prices are not included explicitly in the trip equation. Instead, the economic scarcity of quota is captured implicitly through aggregate TAC levels and vessel-level allocated harvest.

[3] The wind threshold of 8 m/s corresponds approximately to Beaufort scale 5, at which purse-seine operations become difficult for artisanal vessels. We selected this threshold via AIC comparison across 8, 10, and 12 m/s cut-offs on the trip-equation fit.

Trips, harvest, and prices may be jointly determined within the year, implying potential endogeneity in reduced-form trip regressions. The purpose of Eq. (4), however, is not causal identification but rather to provide an empirically grounded behavioral relationship that maps changes in prices, quotas, and environmental conditions into expected fishing effort under projected climate scenarios.

Projection approach

We assess the impact of climate change on the Chilean small pelagic fishery through two channels: (i) a *direct weather channel*, in which projected changes in wind speed alter the number of bad-weather days that constrain fishing operations and thus enter the negative binomial trip equation; and (ii) an *indirect biomass channel*, in which projected changes in SST and CHL modify each stock's intrinsic productivity $r_{i,t}$ through the climate shifter $(\rho_i^{SST}, \rho_i^{CHL})$ of Eq. (2) and, in turn, the law of motion in Eq. (1).

Climate projections are derived from a six-model CMIP6 ensemble (IPSL-CM6A-LR, GFDL-ESM4, CESM2, CNRM-ESM2-1, UKESM1-0-LL, MPI-ESM1-2-HR) selected to span the CMIP6 equilibrium climate sensitivity distribution, under SSP2-4.5 and SSP5-8.5. We apply the delta method (Burke et al. 2015; Free et al. 2019) independently to each model, preserving observed interannual variability and fine-scale spatial structure from satellite records while imposing the CMIP6 climate-change signal. For each model and variable, the delta is computed between the future climatology and a spliced historical baseline (CMIP6 historical 2000–2014 + SSP2-4.5 2015–2024 to match the estimation window; CESM2 chlorophyll under SSP2-4.5 is spliced from SSP5-8.5 where unavailable). We consider mid-century (2041–2060) and end-of-century (2081–2100) windows. Comparative statics integrate over both the structural-parameter posterior and the cross-model ensemble; the variance decomposition is reported in Online Appendix F for stock productivity and Online Appendix G for fleet-level trips.

For the direct weather channel, monthly wind speed deltas from CMIP6 are applied additively to the daily wind speed observations at each vessel's centre-of-gravity (COG); the number of bad-weather days per year is then recomputed at the projected mean using the same 8 m/s operability threshold as in the historical panel. Because the historical daily wind distribution at vessel COGs sits sufficiently close to the threshold, a small upward shift in the mean displaces a non-negligible mass of the upper tail past it, generating substantially more days exceeding the threshold despite the small absolute change in the mean.

Our framework is positioned as long-run comparative statics, not a year-by-year forecast, in line with the bioeconomic-projection tradition of Kasperski (2015). Because the empirical models are estimated on interannual variability rather than decadal trends, the projections capture the mechanical effect of changed environmental and weather conditions on stock productivity and fleet-level effort, while holding prices, quotas, and adaptive behaviour by fishers and managers fixed at historical levels (Auffhammer 2018). They provide a benchmark for the direction and magnitude of climate impacts under a fixed-institution baseline; relaxations that endogenize price adjustment, quota revision, or vessel-level adaptation are signposted as natural extensions of this posterior.

Results

Identification of climate shifters

The central econometric object of this paper is the vector of stock-specific *climate shifters* $\{\rho_i^{SST}, \rho_i^{CHL}\}_{i=1}^{3}$, which modulate the intrinsic growth rate of each stock as

$$r_{i,t} \;=\; r_i^0 \, \exp\!\big(\, \rho_i^{SST} \, (SST_{t-1} - \overline{SST}) \;+\; \rho_i^{CHL} \, (\log CHL_{t-1} - \overline{\log CHL}) \,\big).$$

These shifters admit a clean economic reading: ρ_i^{SST} is the semi-elasticity of the stock's intrinsic productivity r_i with respect to a sea-surface temperature anomaly of one degree Celsius, and similarly ρ_i^{CHL} is the semi-elasticity with respect to a log-point of chlorophyll-a anomaly (a common proxy for primary productivity at the base of the food web). They are the structural parameters that couple the biological law of motion $B_{i,t+1} = B_{i,t} + g_i(B_{i,t}, X_t) - C_{i,t} + \varepsilon_{i,t}$ to the physical climate, and are therefore the quantities that must be identified for the long-run projections of Section to have meaning as *comparative statics under climate change* rather than as out-of-sample forecasts.

We adopt weakly informative priors on $\rho_i^{SST}, \rho_i^{CHL}$ derived from independent reduced-form stress tests (Appendix A) and let the 2000–2024 biomass series update them. Table 2 reports the posterior means, 90% credible intervals, and the posterior-to-prior standard deviation ratio, which measures how much the data refine each parameter relative to the prior.

Table 2: Posterior identification of climate shifters.

Shifter	Stock	Prior $\mathcal{N}(\mu,\sigma)$	Posterior mean	Posterior 90% CI	$\sigma_{\text{post}}/\sigma_{\text{prior}}$	$\widehat{R}$
ρ^{SST}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-1.06	[-1.79, -0.38]	0.43	1.000
ρ^{SST}	Sardina comun CS	$\mathcal{N}(-2.0, 1.0)$	-2.75	[-3.46, -2.02]	0.44	1.000
ρ^{SST}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	0.42	[-1.25, 2.09]	1.01	1.000
ρ^{CHL}	Anchoveta CS	$\mathcal{N}(-2.3, 1.0)$	-3.64	[-5.01, -2.25]	0.83	1.000
ρ^{CHL}	Sardina comun CS	$\mathcal{N}(2.1, 1.0)$	2.17	[0.96, 3.37]	0.74	1.000
ρ^{CHL}	Jurel CS	$\mathcal{N}(0.0, 1.0)$	-0.06	[-1.72, 1.60]	1.01	1.001
ρ^{ENSO}	Jurel CS	$\mathcal{N}(0.0, 0.5)$	-0.02	[-0.81, 0.80]	0.98	1.000

Note:
Shifters are the semi-elasticities of the intrinsic growth rate r_i with respect to SST anomalies (in °C), log-CHL anomalies, and the basin-scale ENSO Niño 3.4 index, respectively. The ENSO row reports the lag-1 refit of Online Appendix E.4, in which jack mackerel productivity is forced by the basin-scale index in place of local coastal shifters; anchoveta and sardine are not refit and inherit their main-specification posteriors. Priors on coastal shifters are adopted from the stock-specific stress-test results reported in Appendix A; the ENSO prior $\mathcal{N}(0, 0.5)$ is deliberately tighter (see Online Appendix E.4). The posterior-to-prior standard-deviation ratio measures prior-to-posterior updating: values near one indicate the data carry little information, whereas values well below one indicate identification from the likelihood. All R-hat <= 1.01.

The pattern of identification is itself informative and organises the economic interpretation of the paper. Three facts stand out.

First, for the two coastal upwelling stocks—anchoveta and sardina común—both climate shifters are identified with substantial precision. The posterior 90% credible intervals exclude zero for all four parameters, and the standard deviation ratios fall between 0.43 and 0.83, indicating that the 2000–2024 biomass series refines the priors materially. For anchoveta, a one-degree positive SST anomaly is associated with a semi-elasticity of $\rho^{SST}_{\text{anch}} = -1.06$ (CI $[-1.79, -0.38]$); for sardina común, the corresponding semi-elasticity is more than twice as large, $\rho^{SST}_{\text{sard}} = -2.75$ (CI $[-3.46, -2.02]$). Evaluated at one historical standard deviation of the SST anomaly series (estimation-sample $\hat{\sigma}_{SST} = 0.26\,^{\circ}\text{C}$ over 2000–2024), a thermal shock of that magnitude lowers the intrinsic growth rate of anchoveta by approximately 24% ($1 - e^{-1.06 \cdot 0.26}$) and that of sardina by approximately 51% ($1 - e^{-2.75 \cdot 0.26}$) relative to the climatological mean. Under a policy-relevant warming signal of $+1\,^{\circ}\text{C}$, comparable to mid-century CMIP6 SSP2-4.5 projections for the Chilean coastal zone, the implied declines are 65% for anchoveta and 94% for sardina, although $+1\,^{\circ}\text{C}$ lies approximately four historical standard deviations outside the estimation window and these projections therefore rely on the log-linear extrapolation embedded in the shifter function. Either way, the magnitudes are economically first-order for a fishery whose Centro-Sur TAC is in the low hundred-thousand-tonne range. The narrow historical SST variance ($\hat{\sigma}_{SST} = 0.26\,^{\circ}\mathrm{C}$) also explains, mechanically, why the one-step-ahead forecasting test of Appendix B fails to discriminate between the climate-augmented and the climate-free specifications: within the estimation window the annual climate signal is small relative to the stochastic process innovation σ_{proc}, yet accumulates to economically consequential levels only over decadal horizons.

Second, the two coastal stocks respond to chlorophyll-a with **opposite signs**: $\rho^{CHL}_{\text{anch}} = -3.64$ (CI $[-5.01, -2.25]$) versus $\rho^{CHL}_{\text{sard}} = +2.17$ (CI $[+0.96, +3.37]$). At one historical standard deviation of $\log CHL$ ($\hat{\sigma}_{\log CHL} = 0.095$, a primary-productivity anomaly of roughly 10%), a positive CHL shock lowers anchoveta growth by approximately 29% while raising sardina growth by approximately 23%. This is a reduced-form footprint of the well-documented trophic asymmetry between the two species, summarised here for the non-specialist reader. Chlorophyll-a is a satellite-observable proxy for phytoplankton biomass, the primary producers at the base of the marine food web. Pulses of high chlorophyll-a indicate periods of strong primary productivity, typically driven by coastal upwelling that brings nutrient-rich deep water to the surface. Sardina común feeds directly on small phytoplankton and benefits from these pulses, so its productivity rises with chlorophyll-a anomalies. Anchoveta, by contrast, feeds on larger zooplankton whose populations are themselves preyed upon by small phytoplankton-feeders; when the food web is dominated by small primary producers, the zooplankton biomass that anchoveta relies on is suppressed, and anchoveta productivity falls. The opposite-signed semi-elasticities of ρ^{CHL}_{anch} and ρ^{CHL}_{sard} identified in Table 2 are consistent with this trophic asymmetry and are not a statistical artefact. The economic implication is first-order for this paper: the artisanal fleet, which lands both species, is not simply “climate-exposed”; it is exposed to **within-target heterogeneity** in the sign of climate effects. A warming-and-greening regime therefore does not translate into a uniform shock on the artisanal sector, but into a substitution across species that the current quota-allocation rules, indexed by species, cannot fully absorb.

Third, and in marked contrast, **neither SST nor CHL shifters are identified for jack mackerel**. The posterior standard deviations σ_{post} are indistinguishable from the prior scale ($\sigma_{\text{post}}/\sigma_{\text{prior}} \approx 1$), the posterior means lie near zero, and the credible intervals straddle zero in both directions. This is the expected result under the null hypothesis that jack mackerel productivity at the Centro-Sur scale is not locally coupled to the Chilean coastal upwelling regime—which is biologically sensible given its transboundary Southeast Pacific stock structure managed through SPRFMO. Econometrically, the model correctly reports non-identification rather than producing a spurious coefficient, which is a strength of the Bayesian state-space approach with structurally informed priors: the data can either move the prior or, as here, confirm that they are silent about it. This is the econometric dividend of Cowles-style structural identification in fishery climate-impact assessment: the structural posterior reports the absence of local climate signal where a reduced-form benchmark would have produced an artefactually precise but structurally meaningless coefficient, which would then propagate into climate projections as if it were information.

Online Appendix E reports five complementary robustness tests of this null—three nested coastal aggregations, a dual-source extension with a Northern acoustic series, basin-scale ENSO shifters at lags 1 and 2, a joint specification with all three shifters active simultaneously, and an attempted SPRFMO regional assessment integration. All five converge on posterior-to-prior ratios between 0.94 and 1.03, indistinguishable from unity, indicating that the local non-identification is structural rather than an artefact of spatial aggregation, sample size, mis-specified forcing scale, or joint-specification convention. The relevant climate forcing for the Centro-Sur jurel stock operates at margins (behaviour, location, phenology) not captured in the elasticity of annual aggregate productivity, consistent with its transboundary SPRFMO management (Arcos et al. (2001); Peña-Torres et al. (2017)).

Figure 3 visualises the posterior–prior updating as a forest plot.

Identification versus short-run prediction

The identified shifters are structural parameters rather than reduced-form correlations: they enter the law of motion as coefficients on exogenous climate inputs, which licenses their use as projection inputs under climate regimes outside the 2000–2024 estimation sample. Their economic value lies in **long-run comparative statics**: given a projected climate regime $X_t^\star$ drawn from the CMIP6 ensemble, the intrinsic productivity under that regime is $r_i^\star = r_i^0 \exp(\rho_i^{SST} \Delta SST_t + \rho_i^{CHL} \Delta \log CHL_t)$, and steady-state biomass, harvest, and fleet-level effort follow mechanically

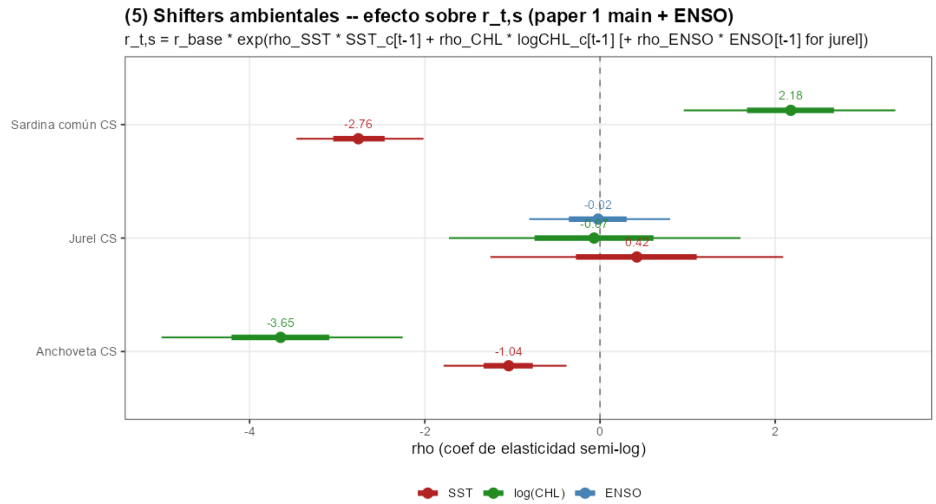

Figure 3: Posterior and prior distributions of the climate shifters $\rho_i^{SST}, \rho_i^{CHL}$ by stock. Anchoveta and sardina común show substantial posterior updating with narrow credible intervals excluding zero; jack mackerel posteriors coincide with the weakly informative prior centred at zero.

(Section). Identification of ρ does not imply superior short-run prediction: a leave-future-out cross-validation at the one-year horizon delivers $\Delta\widehat{\text{ELPD}}_{\text{LFO}} \approx 0.4$ over a climate-free baseline (Online Appendix B), as expected when process noise dominates annual climate innovations. The PSIS-LOO comparison in Online Appendix B confirms the in-sample preference for the climate-augmented specification ($\Delta\widehat{\text{ELPD}}_{\text{LOO}} = 14.59$, SE $= 4.93$); the paper relies on identification-through-fit for the economic interpretation.

Posterior-predictive adequacy

A stock-by-stock comparison of the smoothed latent biomass $B_{i,t}$ against the observed SSB series (Online Appendix C, Figure C.1) shows that the identified model reproduces the historical trajectory within a median 90% posterior band width of approximately 20% of the mean. Residuals are roughly Gaussian and contain no detectable autocorrelation at one-year lag (Online Appendix C, Figure C.2). Posterior-predictive diagnostics are reported in Online Appendix C; Markov-chain convergence diagnostics for all top-level parameters are reported in Online Appendix D.

Total annual trips

Table 3 reports the negative binomial estimates separately for each fleet segment. The specification includes year fixed effects, which absorb aggregate shocks affecting both fleets contemporaneously—most notably the 2019 social outbreak (*estallido social*) and the 2020–2022 COVID-19 period—and identify the structural coefficients β_H (allocated harvest) and β_{weather} (bad-weather days) from within-year cross-vessel variation. Without year fixed effects, $\hat{\beta}^{\text{ART}}_{\text{weather}}$ is approximately twice as large in magnitude, reflecting the contemporaneous correlation between high-wind years and the social/pandemic shocks. The year-FE specification is therefore the basis for the climate projections in Section *Climate change projections*; the relative trip projection $T_v^{\text{FUT}}/T_v^{\text{HIST}}$ does not depend on the year-FE values themselves, which cancel in the ratio.

The coefficients on species-specific allocated harvest reveal cross-fleet heterogeneity that the legacy scalar specification could not capture. The industrial fleet shows a positive and significant response of trips to allocated anchoveta and sardine quotas (quota-binding behaviour), while the response to jack mackerel quota is indistinguishable from zero. The artisanal fleet exhibits a Simpson-paradox pattern in the scalar specification, which the Kasperski-aligned interaction $H^{\text{alloc}}_{\text{anch}} \times$ TIPO_EMB resolves: smaller vessel categories (BM = bote motor, 8%) exhibit a strong positive interaction with anchoveta allocation while larger categories do not. The species-specific quota-share mechanism translates regional TACs into vessel-level effort adjustments under the projections of Section *Climate change projections*.

The two fleets differ markedly in price responsiveness. For the industrial fleet, the jack mackerel price is positive and significant ($p < 0.01$), consistent with this fleet's primary orientation toward jack mackerel. Sardine and anchoveta prices have no detectable effect on industrial effort, reflecting the limited participation of industrial vessels in the sardine–anchoveta fishery. For the artisanal fleet, the sardine price is positive and significant ($p < 0.01$), consistent with sardine being the dominant target species for artisanal vessels in the Centro-Sur region. The anchoveta-price coefficient is negative and significant; this counterintuitive sign likely reflects simultaneity: years of low anchoveta availability generate both higher prices (through the inverse-demand channel) and fewer trips (through reduced catch opportunities), producing a negative reduced-form correlation. As noted above, the purpose of this equation is not causal identification but rather to provide an

Table 3: Negative binomial estimates of annual fishing trips (2013–2024, $N_{IND} = 319$, $N_{ART} = 4201$)

	Dependent variable:	
	Industrial	Artisanal
	(1)	(2)
Hold capacity (log m^3)	0.097 (0.221)	0.132*** (0.046)
Allocated harvest anchoveta ($H^{alloc}_{vy,anch}$, tons)	−0.0003 (0.0002)	0.016*** (0.003)
Allocated harvest sardine ($H^{alloc}_{vy,sard}$, tons)	0.0001*** (0.00003)	0.001*** (0.00004)
Allocated harvest jack mackerel ($H^{alloc}_{vy,jur}$, tons)	0.00002* (0.00001)	0.001 (0.001)
Price anchoveta (1000s $/ton)	−0.004 (0.004)	0.011*** (0.003)
Price sardine (1000s $/ton)	−0.016*** (0.003)	0.003** (0.002)
Price jack mackerel (1000s $/ton)	0.008*** (0.003)	0.007*** (0.002)
Bad weather days	−0.0003 (0.001)	−0.001 (0.0005)
Veda days	−0.079*** (0.005)	0.013*** (0.002)
Vessel type (UNK)	−1.450*** (0.163)	0.160 (0.099)
Vessel type (BR)		−0.329*** (0.079)
Vessel type (BRV)		−0.774*** (0.079)
Vessel type (L)		−2.212*** (0.377)
Vessel type (LM)		0.289*** (0.098)
$H^{alloc}_{vy,anch}\times$ TIPO_EMB (BR)		−0.021*** (0.008)
$H^{alloc}_{vy,anch}\times$ TIPO_EMB (BRV)		−66.546*** (13.119)
$H^{alloc}_{vy,anch}\times$ TIPO_EMB (L)		501.903 (501.942)
$H^{alloc}_{vy,anch}\times$ TIPO_EMB (LM)		−0.017*** (0.003)
$H^{alloc}_{vy,anch}\times$ TIPO_EMB (UNK)		−0.013*** (0.003)
Constant	15.763*** (2.605)	−3.697*** (0.519)
Year fixed effects	Yes	Yes
Pseudo-R^2 (Cragg-Uhler)	0.302	0.631
Observations	319	4,201

Note: *p<0.1; **p<0.05; ***p<0.01.
Vessel-clustered robust SEs. Prices in constant 2018 thousands of pesos/ton, merged at $(year, vessel, region)$ resolution with fallback to year-level national mean when the regional cell is missing.
Allocated harvest is species-specific: $H^{alloc}_{vy,s} = \omega^{hist}_{vs}\bar{Q}^{r(v)}_{sy}$ following Kasperski (2015) Eq. 17. The artisanal regression interacts $H^{alloc}_{vy,anchoveta}$ with TIPO_EMB; TIPO_EMB categories with $N < 50$ vessel-years (BR, BRV, L) are retained in the fit for completeness but excluded from the aggregate factor_trips projections.
Year fixed effects absorb aggregate shocks (estallido social 2019, COVID 2020–2022).
LR test NB vs Poisson: Industrial $\chi^2 = 397.6$; Artisanal $\chi^2 = 37046.9$ (both $p < 0.001$).

empirically grounded mapping for the simulation, in which prices are determined endogenously by the inverse-demand module.

Hold capacity is not significant for the industrial fleet, where vessels are relatively homogeneous in scale, but it is positive and significant for the artisanal fleet. Among artisanal vessels, larger hold capacity is associated with more annual trips, consistent with larger vessels being more commercially active and better able to sustain operations across varying conditions.

The environmental and regulatory variables show distinct patterns. Adverse weather days carry a negative coefficient for the artisanal fleet (vessels in grids with more days of wind above 8 m/s undertake fewer trips, controlling for year-specific aggregate conditions) and is statistically indistinguishable from zero for the industrial fleet, consistent with size-dependent weather tolerance. Biological closure days are strongly negative for the industrial fleet; the artisanal coefficient is positive but does not admit a causal reading under year fixed effects (it identifies only off the cross-zone differential between Valparaíso–Biobío and Araucanía–Los Lagos, conflating locational heterogeneity in artisanal effort intensity with the causal effect of closures). Constructing a year-by-year closure variable that exploits within-zone temporal variation requires reconciling the nominal regulatory ceilings of the D.Ex.~115/1998 and 530/2016 chain with the realised closure days reported in the SUBPESCA Indicadores Biológicos portal from 2019 onwards, which we leave to follow-up work.

Vessel-type dummies capture residual heterogeneity in fleet composition. Among artisanal vessels, larger vessel categories (L, BRV) are associated with substantially fewer trips relative to the baseline, reflecting differences in trip duration and operational patterns across vessel classes.

Climate change projections

The projection methodology is summarized in Section *Projection approach*. Under each scenario and time window, we translate CMIP6-derived environmental deltas into (i) changes in bad-weather days that enter the trip equation (direct channel) and (ii) posterior draws of the stock-specific effective growth rate $r^{\star}_{i,t}$, obtained by evaluating the identified climate shifter of Eq. (2) at the projected SST and log-CHL anomalies (indirect channel).

Projected environmental changes

Table 4 summarises the projected environmental changes as cross-model medians with inter-model IQR. SST warms by +0.7 to +2.4 °C across scenarios. The chlorophyll-a signal is small in magnitude relative to cross-model spread, reflecting documented CMIP6 disagreement on primary-productivity responses. Wind speed increases modestly (+0.1 to +0.4 m/s cross-model median), but translates into +10 to +23.5 additional bad-weather days per year because the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that a small upward shift in the mean displaces a non-negligible mass of the upper tail past it.

Table 4: Projected environmental changes for Centro-Sur Chile (CMIP6 six-model ensemble, delta method; cross-model median with inter-model IQR in brackets).

SSP	Window	Δ SST (°C)	Δ CHL (%)	Δ wind speed (m/s)
SSP2-4.5	2081–2100	+1.44 [+0.96, +1.91]	+0.0 [−3.6, +1.4]	+0.19 [+0.10, +0.24]
SSP2-4.5	2041–2060	+0.74 [+0.52, +0.92]	+2.2 [+0.0, +2.8]	+0.14 [+0.13, +0.15]
SSP5-8.5	2081–2100	+2.39 [+1.93, +3.29]	−1.3 [−7.4, +3.0]	+0.42 [+0.36, +0.45]
SSP5-8.5	2041–2060	+0.92 [+0.61, +1.23]	+0.7 [−2.3, +3.4]	+0.21 [+0.14, +0.32]

Comparative statics on intrinsic stock productivity

Table 5 reports the comparative-statics object $\Delta r_i^\star / r_i^0$, obtained by evaluating the identified climate shifter at the CMIP6 ensemble deltas and integrating over the posterior of $(\rho_i^{SST}, \rho_i^{CHL})$. We report two distinct uncertainty axes separately: the cross-model median and IQR of the model-level posterior medians (climate uncertainty), and the median across models of the within-model 90% posterior credible interval (structural uncertainty); a separate variance decomposition appears in Online Appendix F. Jack mackerel is reported as non-identified ("n.i.' ') consistent with the identification result in Section .

The magnitudes are economically first-order for a Centro-Sur fishery whose combined TAC is in the low hundred-thousand-tonne range. Under the moderate SSP2-4.5 mid-century scenario (cross-model median +0.74 °C; $\Delta \log \mathrm{CHL}$ near zero in the cross-model median, with non-trivial inter-model spread), the cross-model median of anchoveta's intrinsic productivity declines by 51% and sardine's by 79%, with posterior probability of negative impact equal to one in both cases. Under the high-emissions SSP5-8.5 end-of-century scenario (cross-model median +2.39 °C, $\Delta \log \mathrm{CHL}$ cross-

Table 5: Comparative statics on intrinsic stock productivity under the CMIP6 six-model ensemble.

Stock	Scenario	n_m	ΔSST (°C)	$\Delta \log$ CHL	$\Delta r_i^*/r_i^0$ (cross median)	Cross-model IQR	Within posterior 90% CI	$\Pr(\Delta < 0)$
Anchoveta CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-50.7%	[-52.7%, -47.8%]	[-68.0%, -27.0%]	1.00
Anchoveta CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-71.0%	[-76.3%, -60.4%]	[-91.1%, -31.0%]	1.00
Anchoveta CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-62.4%	[-68.0%, -55.9%]	[-80.9%, -31.2%]	1.00
Anchoveta CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-89.6%	[-91.5%, -87.4%]	[-98.3%, -54.2%]	0.99
Sardine CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	-78.6%	[-93.2%, -72.5%]	[-85.7%, -67.3%]	1.00
Sardine CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	-97.2%	[-98.9%, -89.8%]	[-98.9%, -92.6%]	1.00
Sardine CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	-91.1%	[-96.7%, -78.5%]	[-95.1%, -83.2%]	1.00
Sardine CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	-99.9%	[-100.0%, -99.4%]	[-100.0%, -99.3%]	1.00
Jack mackerel CS	SSP2-4.5, 2041-2060	5	+0.58	+0.022	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP2-4.5, 2081-2100	5	+1.31	+0.000	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2041-2060	6	+0.92	+0.007	n.i.	n.i.	n.i.	n.i.
Jack mackerel CS	SSP5-8.5, 2081-2100	6	+2.39	-0.013	n.i.	n.i.	n.i.	n.i.

Note:
Entries summarise the posterior of the proportional change in intrinsic productivity, Delta r / r0 = exp(rho_SST * Delta SST + rho_CHL * Delta log CHL) - 1, evaluated at scenario-window-specific CMIP6 anomalies for each of n_m models in the ensemble and integrated over the posterior of (rho_SST, rho_CHL) from the full state-space specification. The cross-model IQR is the 25th-to-75th percentile range across CMIP6 models of the within-model posterior medians and captures inter-model climate uncertainty. The within posterior 90% CI is the cross-model median of the within-model 5th-to-95th-percentile posterior interval and captures structural uncertainty about (rho_SST, rho_CHL) holding the climate forcing fixed. Jack mackerel CS is reported as non-identified ('n.i.') because the posterior of its shifters coincides with the weakly informative prior; see the identification discussion in Results. n_m = 5 instead of 6 in the SSP2-4.5 panels because CESM2 does not publish chl-ssp245 in the NCAR CMIP6 catalogue. Under SSP5-8.5 end-of-century the implied SST anomaly lies approximately nine in-sample standard deviations outside the realised 2000-2024 envelope, so it is best read as a structural-form stress test under the maintained log-linear shifter. Raw numerical values are archived in tables/growth_comparative_statics_raw.csv; model-level disaggregation is in tables/growth_comparative_statics_by_model.csv. Appendix F reports the variance decomposition between cross-model and within-model components.

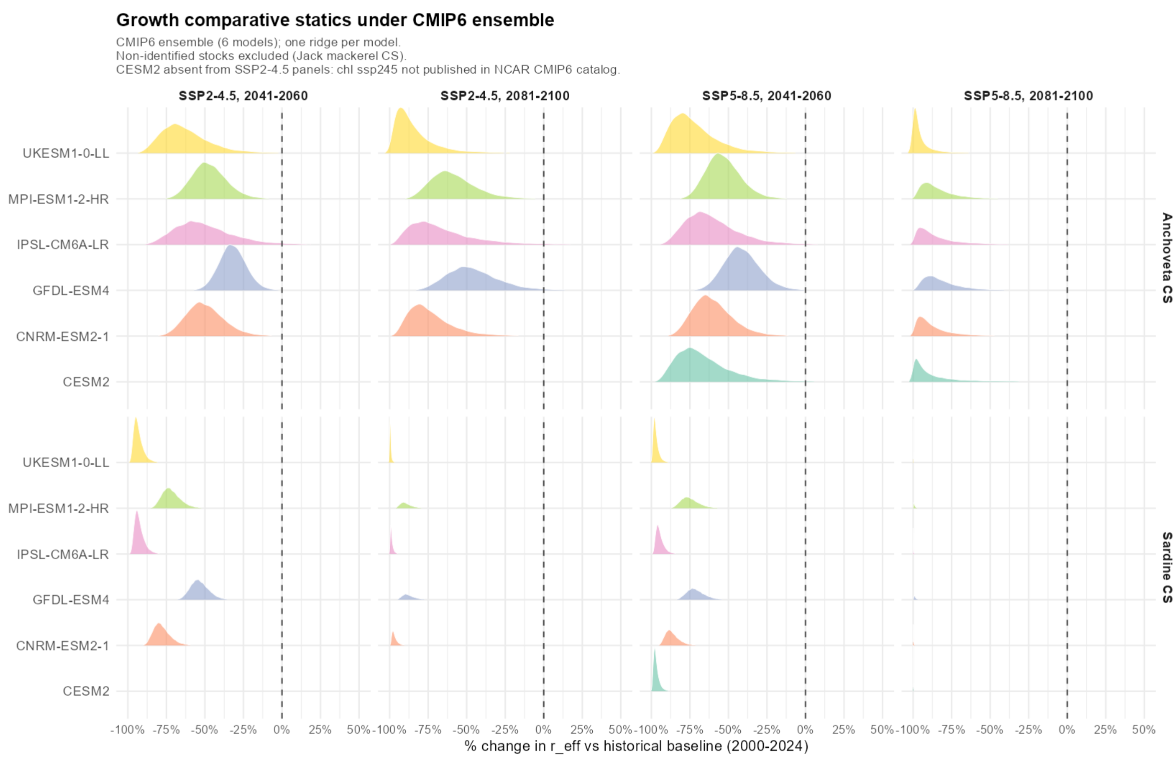

Figure 4: Posterior distribution of the proportional change in intrinsic stock productivity under CMIP6 scenarios, six-model ensemble. Ridges are kernel-smoothed marginal posteriors of the proportional change in intrinsic productivity derived from draws of the state-space full specification, with one ridge per CMIP6 model in each scenario–window pair; vertical line at zero. CESM2 is absent from the SSP2-4.5 panels because chl-ssp245 is not published in the NCAR CMIP6 catalogue. Jack mackerel is omitted because its climate shifter is not identified in the 2000–2024 sample.

model median -0.013 with 5th-percentile of -0.30 across models) the cross-model medians deepen to -90% and -99.9% respectively, and the entire 90% credible interval of sardine's response lies below zero in every model of the ensemble; the anchoveta cross-model probability of decline is 0.99 rather than 1 because under a small fraction of (model, draw) pairs a sufficiently negative $\Delta \log \mathrm{CHL}$ combined with $\rho_{\text{anch}}^{CHL} < 0$ offsets the SST-driven collapse. The identified signs confirm the qualitative reading of the shifters in Section : both coastal stocks respond negatively to SST, but sardine carries roughly twice the SST semi-elasticity of anchoveta, and its CHL semi-elasticity is of the opposite sign, so that the compound response—negative thermal shock combined with negative primary-productivity shock—is sharper for sardina than for anchoveta.

The comparative-statics exercise is a long-run object rather than a year-by-year forecast. It describes the steady-state intrinsic productivity of each stock under a counterfactual climate, *holding fixed* the law of motion and the fishery's regulatory regime. The shifters $(\rho_i^{SST}, \rho_i^{CHL})$ enter the law of motion as coefficients on exogenous climate forcings, which makes them invariant to the joint distribution of (SST, B, C) and therefore evaluable at climate regimes outside the estimation window—in particular, at an SST anomaly of $+2.3°\mathrm{C}$ that lies roughly nine historical standard deviations beyond the 2000–2024 mean. A purely correlational specification would not license this extrapolation.

Implications for fleet-level effort

Table 6 translates the comparative-statics object on intrinsic stock productivity in Table 5 into a comparative-statics object on fleet-level annual trips, by propagating each posterior draw through the stock-dynamics steady-state equation under historical average fishing pressure and through the negative binomial trip equation of Section *Total annual trips*. Vessel-level responses are aggregated to the fleet level as medians over posterior draws and over vessels within fleet, weighted by each vessel's realized catch composition over the 2012–2024 estimation window.

The posterior probability of portfolio loss—defined as a reduction of more than fifty percent in the composition-weighted biomass factor f_v^H relative to the 2000–2024 historical baseline—exceeds 0.95 for the artisanal fleet under every CMIP6 scenario considered and reaches 0.99 under SSP5-8.5 end-of-century. For the industrial fleet, the loss probability is a stable 0.12 across all four

Table 6: Long-run implications of CMIP6 climate scenarios for fleet-level fishing effort, under a Schaefer steady-state thought experiment with fishing pressure fixed at the 2000–2024 historical average. Entries are cross-model medians with cross-model IQRs in brackets, aggregated within each CMIP6 model over posterior draws and vessels within fleet.

Fleet	Scenario	$\Pr(f_v^H < 0.5)$	%Δ trips (marginal)	%Δ trips (conditional)
Artisanal	SSP2-4.5, 2041-2060	0.95 [0.93, 0.96]	-13.2% [-13.3%, -12.2%]	-4.6% [-4.8%, -4.0%]
Artisanal	SSP2-4.5, 2081-2100	0.98 [0.97, 0.99]	-14.1% [-14.3%, -13.7%]	-3.9% [-4.5%, -3.4%]
Artisanal	SSP5-8.5, 2041-2060	0.98 [0.96, 0.98]	-13.6% [-13.8%, -13.4%]	-4.2% [-4.5%, -4.1%]
Artisanal	SSP5-8.5, 2081-2100	0.99 [0.99, 0.99]	-15.0% [-15.2%, -14.9%]	-3.8% [-5.5%, -2.6%]
Industrial	SSP2-4.5, 2041-2060	0.12 [0.11, 0.12]	-1.0% [-1.0%, -0.8%]	-0.7% [-0.7%, -0.7%]
Industrial	SSP2-4.5, 2081-2100	0.12 [0.12, 0.12]	-0.9% [-1.0%, -0.8%]	-0.7% [-0.8%, -0.7%]
Industrial	SSP5-8.5, 2041-2060	0.12 [0.12, 0.12]	-0.8% [-0.9%, -0.8%]	-0.7% [-0.9%, -0.7%]
Industrial	SSP5-8.5, 2081-2100	0.12 [0.12, 0.12]	-1.0% [-1.1%, -1.0%]	-1.0% [-1.0%, -0.9%]

Note:
Entries summarise the posterior of the proportional change in fleet-level annual trips, computed for each of the six CMIP6 models in the ensemble and then aggregated across models. Within each model, the posterior is integrated over draws of the full state-space specification and vessels within fleet (composition-weighted exposure). Cross-model medians and 25th-to-75th-percentile inter-quartile ranges across the n_m models are reported in brackets; the median across models of the within-model 5th-to-95th-percentile posterior interval (within posterior CI) is archived in tables/trip_comparative_statics_raw.csv. f_H is the composition-weighted biomass factor relative to the historical baseline: f_H = sum over species of omega_vs * B_star_s / B_hist_s, where B_star_s = K_s * (1 - F_hist_s / r_eff_s) is the Schaefer steady-state biomass and omega_vs is the vessel's realized catch share in species s. Pr(f_H < 0.5) reports the posterior probability that the vessel's composition-weighted biomass factor falls below one-half. The marginal median integrates over all posterior draws within each model; the conditional median restricts to draws with f_H >= 0.5, isolating the pure climate-elasticity component from the steady-state collapse component of the trip response. Jack mackerel is treated as non-identified (factor_B = 1) consistent with the identification discussion in the Results section. Raw numerical values, by-model disaggregation, and by-stock extinction probabilities are archived in tables/trip_comparative_statics_raw.csv, tables/trip_comparative_statics_by_model.csv, and tables/trip_comparative_statics_extinct.csv. The narrow cross-IQR observed in many cells reflects a floor effect: once Schaefer steady-state collapse saturates the portfolio-weighted biomass factor across models, the trip response converges to exp(-beta * H_alloc * 1), independent of climate magnitude. Appendix G reports the formal two-way variance decomposition of the fleet-level trip response, which confirms that the within-model component (posterior plus vessel heterogeneity) dominates the total variance under saturated regimes; Appendix F reports the analogous decomposition for the underlying productivity response.

scenarios, driven entirely by its five-percent exposure to sardine; the remaining 95% of its historical portfolio is allocated to jack mackerel, whose climate shifter is treated as non-identified in Section . The asymmetry in loss probability between the two fleets is first-order and stable across climate pathways, approximately eight-fold across all four scenarios.

Decomposing the fleet-level trip response isolates the channels through which this asymmetry operates. The *marginal* posterior median of %Δ trips ranges from −13.2% (SSP2-4.5 mid) to −15.0% (SSP5-8.5 end) for the artisanal fleet and from −0.8% to −1.0% for the industrial fleet—a ratio of roughly fifteen to one at end-of-century. The *conditional* posterior median, restricted to draws with $f_v^H \geq 0.5$, isolates the climate-elasticity component from the portfolio-collapse component and yields -3.8% to -4.6% for artisanal and -0.7% to -1.0% for industrial, a conditional ratio of roughly four to one. Two channels generate the asymmetry. The *indirect biomass channel* transmits the climate signal asymmetrically through portfolio composition: the artisanal fleet is concentrated in the two coastal-upwelling stocks with identified climate shifters, while 95% of the industrial fleet's portfolio sits in jack mackerel, whose Centro-Sur shifter is non-identified and held at the climatological mean. The *direct weather channel* transmits the signal through differential weather sensitivity: $\hat{\beta}^{\text{ART}}_{\text{weather}} = -0.0007$ (marginally significant) vs $\hat{\beta}^{\text{IND}}_{\text{weather}} = -0.0003$ (indistinguishable from zero), translating the projected +10 to +23.5 days of additional bad weather into a -0.7% to -1.9% contribution to the artisanal response and near-zero for industrial. The limited cross-sector transferability of the prevailing sector-level allocation locks this dual exposure into differential losses of harvest capacity, consistent with the portfolio mechanism documented by Kasperski and Holland (2013) and the access constraints of Oken et al. (2021).

The between-fleet asymmetry decomposes further into within-artisanal heterogeneity by vessel type (Table 7). The dominant *lancha motor* (LM) segment (77% of the panel) contracts by -19.9% under SSP5-8.5 end-of-century, driven by its identified positive sardine elasticity and the projected -99.9% steady-state collapse of the sardine stock. The smaller *bote motor* (BM) segment (8%) contracts by only -3.2% at the marginal median, because its identified positive anchoveta elasticity ($\hat{\beta}_h^{\text{anch,BM}} = +0.016$, $p < 0.001$) partially offsets the sardine contraction in the high-mass posterior region. The intermediate UNK segment (14%) contracts by -7.6%. Portfolio-loss probability exceeds 0.98 in all three categories, so the differential within artisanal is concentrated in the magnitude rather than the incidence of portfolio collapse.

Three caveats qualify this thought experiment. First, the Schaefer steady-state comparative statics holds fishing pressure fixed at the 2000–2024 historical average; any regulatory response that lowered F in the face of falling productivity would attenuate both the loss probability and the marginal trip response reported here. Second, the 2000–2024 window was not itself in Schaefer equilibrium under F_{hist}: an internal consistency check evaluated at $\Delta SST = 0$ and $\Delta \log CHL = 0$ yields a median f_v^H of 1.015 rather than 1.000, so the results should be read as responses *relative to the baseline steady-state*, not relative to the observed 2000–2024 trajectory. Third, jack mackerel

Table 7: Within-artisanal-fleet decomposition of the long-run trip response by vessel type, under SSP5-8.5 end-of-century. Entries are cross-model medians with cross-model IQRs in brackets. The marginal column integrates over all posterior draws; the conditional column restricts to draws in which the composition-weighted biomass factor is at least one-half, isolating the climate-elasticity component from the steady-state collapse component.

	Segment	Pr(portfolio loss)	%Δ trips (marginal)	Cross-model IQR	%Δ trips (conditional)
LM	Lancha motor (LM)	0.99	-19.9%	[-20.0%, -19.7%]	-16.1%
UNK	Unidentified (UNK)	0.98	-7.6%	[-8.1%, -7.3%]	-1.6%
BM	Bote motor (BM)	1.00	-3.2%	[-3.3%, -3.0%]	+16.3%

Note:
Categories with fewer than 50 vessel-year observations (BR, BRV, L) collectively account for less than 0.1 percent of artisanal effort and quota and are excluded from this table. Vessel type BM is the reference category in the negative binomial interaction; LM and UNK enter as deviations. Raw cross-model values and other scenario-window combinations are archived in trip_comparative_statics_by_tipo_emb_raw.csv.

is treated as non-identified at the Centro-Sur scale, a null on *local identification* rather than on biological climate sensitivity: the 25-year window provides only 16 informative annual observations against a historical SST envelope of $\hat{\sigma}_{SST} \approx 0.26\,^{\circ}\mathrm{C}$. Five complementary tests reported in Online Appendix E (nested coastal domains, dual-source extension, basin-scale ENSO, joint specification, SPRFMO integration) all return posterior-to-prior ratios indistinguishable from unity, indicating that the local null is structural. The industrial fleet's apparent climate protection in Table 6 therefore rests on a null about local identification, which a range-wide SPRFMO analysis would be the appropriate setting to refine.

Discussion

Our results reveal a substantial asymmetry in how climate change affects different segments of Chile's small pelagic fishery, governed by the differential exposure of each fleet's species portfolio to the identified climate parameters under limited cross-sector transferability. The artisanal fleet, concentrated in the sardina–anchoveta pair, is the segment whose long-run harvest capacity is most exposed to the CMIP6 signal; the industrial fleet's 95% exposure to jack mackerel (whose local productivity is non-identified) buffers it under the maintained convention. This finding aligns with the broader literature on heterogeneous climate impacts within fleets (Sumaila et al. 2011; Free et al. 2019) and underscores that aggregate projections can mask first-order distributional consequences inside the same fishery. The asymmetry further decomposes within the artisanal fleet by vessel type: the dominant LM segment contracts by approximately twenty percent, while the smaller BM segment contracts by only three to four percent (Table 7).

The non-identification of $(\rho_{\text{jur}}^{SST}, \rho_{\text{jur}}^{CHL})$ at the Centro-Sur scale is itself a substantive result rather than a nuisance, but its policy reading depends on the distinction between *non-identification* and *no climate effect*. Five converging tests reported in Online Appendix E (varying coastal aggregation, augmenting biomass coverage, replacing local shifters with basin-scale ENSO, joint specification, and SPRFMO integration) all return posterior-to-prior ratios in 0.94–1.03. Existing empirical evidence documents that climate forcing operates on jack mackerel at scales broader than local coastal anomalies-\/-\/-@Arcos2001-jq on intrusion of the 15 °C isotherm during the 1997–98 El Niño, Peña-Torres et al. (2017) on ENSO-driven shifts in location choices in the Chilean fishery—but each documents climatic effects on *margins* (location choice, range distribution, spawning timing) that are not necessarily captured in the elasticity of annual aggregate productivity. Closing the identification gap will require either a longer biomass record or a basin-scale SPRFMO assessment with explicit cross-stock spillover. We accordingly hold $r^*_{\text{jack mackerel}}$ fixed in the projections; a prior-propagation sensitivity reported in Online Appendix E.4 returns a 90% predictive interval of approximately three orders of magnitude on the productivity factor under SSP5-8.5 end-of-century, confirming that the fixed-r^* convention is the defensible reporting choice.

The two channels through which climate change reaches fleet-level effort operate on very different scales and on different segments of the fleet. The direct weather channel is small in mean wind anomaly—the cross-model median CMIP6 increase in near-surface wind speed is below +0.5 m/s even under SSP5-8.5 end-of-century—but the historical daily wind distribution at vessel COGs sits sufficiently close to the 8 m/s operability threshold that additional bad-weather days exceed twenty per year on average under SSP5-8.5 end-of-century. This translates into a −0.7% to −1.9% marginal contribution to the artisanal trip response and a contribution indistinguishable from zero for the industrial fleet, reflecting the differential weather sensitivity captured by $\hat{\beta}_{\text{weather}}$. The indirect biomass channel dominates the response in absolute terms for both fleets but operates entirely through portfolio composition, so the asymmetry it generates is intrinsic to the fixed sector-level allocation of stocks across segments. Climate adaptation policy in the Centro-Sur therefore needs to address both margins. The indirect channel calls for reform of the institutional architecture that mediates portfolio exposure—quota allocation rules, cross-fleet transferability, and the design of TACs under shifting species productivity. The direct channel calls for operational measures targeted at the artisanal segment specifically, such as expanded port infrastructure, vessel reinforcement programs, or weather-indexed insurance, which are first-order for that fleet but redundant for the industrial purse-seine fleet. This conclusion is based on the CMIP6 monthly wind ensemble; downscaled regional climate models that resolve extreme wind events could only raise the weight of the direct channel further.

CMIP6 SST anomalies under SSP5-8.5 end-of-century reach roughly $+2.3\,^{\circ}\mathrm{C}$, nearly an order of magnitude outside the 25-year estimation window ($\hat{\sigma}_{SST} \approx 0.26\,^{\circ}\mathrm{C}$). The corresponding posterior of $\Delta r_i^{\star}/r_i^0$ therefore leans on the log-linear extrapolation embedded in the shifter function. Thermal-tolerance arguments in the climate-fisheries literature (Cheung et al. 2010; Free et al. 2019) imply that for stocks operating below their thermal optimum, a log-linear specification, if anything, *understates* the productivity response under high warming, so the projected magnitudes should be read as conservative within the class of monotone shifter functions consistent with the identified posterior signs.

Seven caveats qualify the present analysis: holding ex-vessel prices and management rules constant; substantial cross-model spread in chlorophyll-a; identification on a 25-year series limiting non-linearity detection; the absence of risk preferences in the trip equation; the zonal coding of biological closures; the purse-seine focus of the trip-equation panel; and the missing wind components for CESM2 in the direct weather channel. Online Appendix I states each in full and qualifies its implication for the interpretation developed above.

The finding that climate change creates winners and losers within the same fishery has direct implications for quota-allocation policy. A natural first reading would attribute the asymmetry to rigid single-species permits in the artisanal sector, but both sectors operate *de jure* multi-species permit portfolios: SUBPESCA's 2012 registry records simultaneous authorisations across all three pelagic species for 76% of industrial vessels (Subsecretaría de Pesca y Acuicultura (SUBPESCA) 2026); the IFOP logbook in turn reveals that 90% of artisanal vessels operate a multi-species portfolio in any given year. The asymmetric climate sensitivity we identify is therefore not driven by permit count but by four dimensions of the institutional architecture. First, the statutory partition fixes inter-sectoral shares at the legislative level (Table 1). Second, the partition admits a single administratively rationed re-allocation channel: industrial titulares cede 85–99% of their assigned anchoveta and sardine TACs to artisanal cessionaries each year (25–77 kt per species-year), while jack mackerel cessions are an order of magnitude smaller and reverse direction across years (Online Appendix H.3). The channel is non-market in the conventional sense but operational rather than dormant. Third, within-sector transferability is asymmetric: industrial coefficients trade under LTP since 2013, while artisanal RAE assignments remain locked to the regional registry. Fourth, the revealed portfolio composition differs in its climate correlation structure: the artisanal logbook shows 80–86% of activity concentrated in the anchoveta–sardina pair (both with identified negative climate elasticities), while 67% of industrial activity sits in the same pair with the remainder in jack mackerel and caballa (non-identified at the Centro-Sur scale). The asymmetry the paper estimates therefore quantifies the cost of a statutory partition combined with a climate-correlated artisanal portfolio under projected SSP5-8.5 stress. Online Appendix H reports the descriptive evidence and Online Appendix H.2 computes a policy counterfactual.

Implications of the 2025 fractioning reform

The asymmetry estimated in this paper is identified on the pre-reform fractioning regime that prevailed during the estimation window (2013–2024). The 2025 reform, in force from 2026 until 2040, modifies that regime in two relevant directions for the Central-South small pelagic fishery (Congreso Nacional de Chile 2025). First, the artisanal share of the anchoveta and sardine TACs rises from 78% to 90% across the Valparaíso–Los Lagos macrozone; the same artisanal sector therefore now holds a deeper claim on stocks whose long-run productivity is projected to collapse under SSP5-8.5 end-of-century. Second, the artisanal share of the jack mackerel TAC rises from 10% to 30% in Valparaíso–Los Ríos and from 10% to 15% in Los Lagos—a quota-weighted aggregate of approximately 28% across the macrozone—giving the artisanal sector partial access to the species whose local productivity is non-identified at the Centro-Sur scale and whose climate response is held fixed in our projections. Table 8 reports the static plug-in counterfactual at SSP5-8.5 end-of-century, holding the structural posterior and the TAC level at 2024 values and varying only the statutory artisanal share.

Table 8: Static counterfactual: artisanal absolute landing under the pre-reform and 2025 fractioning regimes at SSP5-8.5 end-of-century, Centro-Sur (Valparaíso–Los Lagos), 2024 Cuotas Globales.

		Artisanal share (α_s)		Artisanal landing L_{art} (kt	
Stock	TAC 2024 (kt)	Pre-reform	2025 reform	Pre-reform	2025 reforn
Anchoveta (Valparaíso–Los Lagos)	211.3	0.78	0.90	17.1	19.
Sardina común (Valparaíso–Los Lagos)	290.6	0.78	0.90	0.2	0.
Jack mackerel (Valparaíso–Los Ríos)	578.6	0.10	0.30	57.9	173.
Jack mackerel (Los Lagos)	80.6	0.10	0.15	8.1	12.
Total Centro-Sur				**83.3**	**205.**

Notes. Static plug-in evaluation of $L_{\text{art}}^{(R)} = \sum_s \alpha_s^{(R)} \cdot (1 + \widehat{\Delta r_s^\star / r_s^0}) \cdot Q_s^0$, with $\widehat{\Delta r_s^\star / r_s^0}$ set to the cross-model median at SSP5-8.5 end-century from Table 5 (−89.6% anchoveta, −99.9% sardine, 0% jack mackerel by the non-identification convention of Section). Q_s^0 is t 2024 Cuota Global de Captura. Artisanal fractions follow the pre-reform statutory schedule and the 2025 reform. The Los Ríos–Los Lag jack mackerel quota is reported as a single block by SERNAPESCA; the row applies the Los Lagos artisanal share (15%) as a conservat lower bound on the true sub-block share, which would rise toward 30% to the extent that Los Ríos contributes to the aggregate quo The posterior-propagated version of this table, with cross-model IQR and within-posterior 90% CI, is reported in Online Appendix H

Three readings of Table 8 structure the policy interpretation. First, the redistribution *raises* the absolute artisanal landing under projected SSP5-8.5 end-of-century stress by approximately 122 kt at the Centro-Sur aggregate, almost entirely through the 20-percentage-point increase in the artisanal jack mackerel share in Valparaíso–Los Ríos rather than through anchoveta and sardine, whose projected productivity collapse makes the 12-percentage-point increase in the artisanal share mechanically small. Second, the policy logic of the redistribution—designed to protect the artisanal sector—in fact *deepens* the artisanal exposure to anchoveta and sardine biomass collapse under climate stress: the same sector now holds 90% rather than 78% of a TAC projected to fall to floor levels. The net welfare effect therefore depends on whether the jack mackerel access expansion outweighs the deepened anchoveta-sardine exposure. Third, the counterfactual is a de jure upper bound on the post-cession allocation. Under the pre-2025 regime the industrial sector ceded 85–99% of its assigned anchoveta and sardine TACs to artisanal cessionaries via the cession channel (Online Appendix H.3), so the 12-percentage-point rise in the artisanal de jure share translates into a much smaller change in realised landings; the operationally meaningful change is the contraction of the cession volume that industrial titulares had been intermediating for fee. The reform is therefore primarily *redistributive* (transferring rent from industrial titulares to artisanal armadores) rather than *allocative* in anchoveta and sardine; for jack mackerel, historical cessions are an order of magnitude smaller and the de jure expansion is operationally meaningful.

Conclusions

This paper identifies the structural climate elasticities of stock productivity in the Chilean small pelagic fishery and combines them with a vessel-level effort equation to compute long-run comparative statics on fleet-level fishing effort under projected climate paths. We contribute in three ways. First, we couple a Bayesian state-space model with negative binomial trip equations estimated separately by sector, showing that the direction of the distributional asymmetry across fleets is governed by the interaction between portfolio composition and the limited cross-sector transferability of the prevailing allocation, extending the portfolio-and-access mechanism of Kasperski and Holland (2013), Cline et al. (2017), and Oken et al. (2021) to a climate-shock setting. Second, we identify $(\rho_i^{SST}, \rho_i^{CHL})$ sharply for the two coastal-upwelling stocks and document non-identification for jack mackerel at the Centro-Sur scale. Third, the variance decomposition shows that once the stock-dynamics steady-state collapse saturates the artisanal portfolio, the within-model component accounts for essentially all variance in projected fleet-level effort, so the policy-relevant uncertainty lies in the structural posterior rather than in CMIP6 disagreement.

The projected comparative statics indicate that the long-run intrinsic productivity of the coastal-upwelling pair—anchoveta and sardine—declines sharply under all CMIP6 scenarios considered. Sardine is consistently more exposed than anchoveta, with the SSP5-8.5 end-of-century scenario driving its steady-state productivity to floor levels at the Centro-Sur scale. The Centro-Sur jack mackerel stock, by contrast, does not provide sufficient information to identify a local climate elasticity in the 2000–2024 sample, consistent with its transboundary SPRFMO management. These results suggest that climate-adaptation policy in this fishery should target the quota-allocation architecture—specifically, the binding industrial–artisanal split and the limited cross-sector transferability that locks each fleet into its historical portfolio—rather than operational margins alone.

Several extensions are natural. A bi-level Stackelberg optimisation that uses our posterior as prior would deliver full forward biomass trajectories under endogenous TACs and effort allocation, following Kasperski (2015) with explicit cost functions and an inverse-demand system. The spatial dimension of effort allocation is a second extension, connecting the multi-species model to the location-choice literature (Dupont 1993; Smith 2005; Hicks et al. 2020); migrating the annual NB trip equation to a daily discrete-choice specification with environmentally informed SDMs as proxies for local availability would capture daily and within-port heterogeneity, following Quezada-Escalona et al. (2026). Finally, incorporating risk preferences and production risk would provide a richer characterisation of fleet responses to projected environmental variability (Kasperski and Holland 2013).

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Data and code availability

All code required to replicate the analysis is openly available at https://github.com/fquezadae/Impact-of-Environmental-Variability-on-Harvest and will be archived at submission as a versioned Zenodo release with a citable DOI. Environmental covariates are obtained from publicly available sources (E.U. Copernicus Marine Service for the historical period; CMIP6 archive via the Earth System Grid Federation for projections), and stock-assessment biomass series from IFOP and SPRFMO annual technical reports. Vessel-level logbook data are subject to restricted-access conditions under the IFOP–Convenio Desempeño framework; vessel identifiers have been anonymised, and aggregate-level processed data sufficient to reproduce all reported tables and figures are included in the replication repository. Detailed data-provenance information is provided in the cover letter accompanying t

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